

Capstone Project: Azure DevOps CI/CD Pipeline for a Python Application

Business Scenario

A retail company wants to automate its Python application deployment process using Azure DevOps. As part of the modernization initiative, the engineering team has decided to adopt Agile planning and Continuous Integration (CI).

Your team has been assigned the responsibility of planning the work, managing the source code, and implementing a CI pipeline that automatically builds and validates the application whenever changes are committed.

Objective

Complete the following activities using Azure DevOps.

Part 1 – Agile Planning

The Product Owner has provided the following business requirement:

Epic

Develop a Customer Management System for a Retail Company

The application should allow users to manage customer records by adding, viewing, updating, and deleting customer information.

Your Task

Break down the above Epic into:

- Features

- User Stories
- Tasks

Create the complete work hierarchy in Azure DevOps Boards.

Part 2 – Source Code Management

Create a new Git repository in Azure DevOps named:

```
customer-management-system
```

Develop a simple Python application.

Example:

```
class Customer:

    def __init__(self, customer_id, name):
        self.customer_id = customer_id
        self.name = name

    def display(self):
        print(f"Customer ID : {self.customer_id}")
        print(f"Customer Name : {self.name}")

customer = Customer(101, "Rahul Sharma")
customer.display()
```

Commit and push the source code to the Azure DevOps repository.

Part 3 – Continuous Integration (CI)

Create an Azure DevOps YAML pipeline that:

- Retrieves the source code from the repository.
- Sets up a Python environment.
- Installs project dependencies (if any).
- Executes the Python application.
- Verifies that the build completes successfully.

A successful pipeline execution should display the application output in the build logs.

Part 4 – Validation

Verify that:

- The repository contains the application source code.
 - The Azure DevOps Boards contain the Agile work items.
 - The YAML pipeline executes successfully.
 - The build completes without errors.
 - The application output is visible in the pipeline logs.
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Deliverables

Prepare a Document with the screenshots for the work done

- Azure DevOps Project
- Azure DevOps Boards (Epic → Features → User Stories → Tasks)
- Git Repository
- Python Source Code
- Azure Pipelines YAML File
- Successful CI Pipeline Execution